

Semantic Test Document for AI Entity and Relation Extraction

This document is intentionally written for testing a RAG knowledge graph pipeline. It contains clear named entities, repeated canonical names, and explicit semantic relations between AI models, methods, tasks, frameworks, and concepts.

The expected ontology contains five labels: Model, Method, Task, Framework, and Concept. The text avoids references, author names, broken tables, and URLs so that the NER layer can be evaluated on semantic content rather than PDF noise.

1. Transformer-based language models

The Transformer is a neural network architecture based on self-attention. Self-Attention is a method that allows a model to compare tokens inside the same sequence. Attention is a broader concept, while Self-Attention is the specific method used inside Transformer blocks.

BERT is a Model that uses Masked Language Modeling. BERT is used for Named Entity Recognition, Text Classification, and Question Answering. RoBERTa is a Model derived from BERT and also uses Masked Language Modeling for pretraining.

GPT is a Model that uses autoregressive language modeling. GPT is used for Text Generation, Summarization, and Question Answering. GPT-4 is a Model from the GPT family and is often evaluated on reasoning and code generation tasks.

T5 is a Model that treats many NLP problems as Text-to-Text Generation. T5 is used for Machine Translation, Summarization, and Question Answering. BART is a Model used for Summarization and Text Generation.

2. Modern large language models

LLaMA is a Model based on the Transformer architecture. LLaMA uses Rotary Positional Embeddings and the SwiGLU activation function. LLaMA can be adapted with LoRA for parameter-efficient fine-tuning.

Mistral-7B is a Model that uses a Transformer decoder architecture. Mixtral-8x7B is a Model that uses Mixture of Experts. Mixture of Experts is a Method that routes tokens to specialized expert networks.

Qwen is a Model used for Text Generation, Code Generation, and Question Answering. DeepSeek-Coder is a Model used for Code Generation. Whisper is a Model used for Speech Recognition.

CLIP is a Model that uses Contrastive Learning. CLIP connects images and text in a Joint Embedding Space. Contrastive Learning is a Method used for Image Retrieval and Zero-Shot Classification.

3. Adaptation and alignment methods

Fine-tuning is a Method used to adapt a pretrained model to a downstream task. Instruction Tuning is a Method used to make a language model follow natural language instructions. RLHF is a Method used to align a model with human preferences.

LoRA is a Method for parameter-efficient fine-tuning. QLoRA is a Method that combines quantization with LoRA. Quantization is a Method used to reduce memory usage and improve inference efficiency.

Prompt Engineering is a Method used to control model behavior through prompts. Chain-of-Thought Prompting is a Method used for reasoning tasks. Retrieval-Augmented Generation is a Method that combines retrieval with text generation.

4. RAG architecture and vector search

Retrieval-Augmented Generation, also called RAG, is a Method used for Question Answering over external knowledge sources. RAG uses a Vector Index to retrieve relevant chunks before generation.

Embedding is a Concept that represents text as vectors. A Vector Index is a Concept used for semantic search. Semantic Search is a Task that uses embeddings to find relevant documents.

Milvus is a Framework used for vector search. Qdrant is a Framework used for vector search. Weaviate is a Framework used for vector search. LlamaIndex is a Framework used to build RAG applications. LangChain is a Framework used to build LLM applications and agent workflows.

A typical RAG pipeline uses Document Chunking, Embedding, Vector Search, Context Retrieval, and Answer Generation. The Retriever component uses Embedding and Vector Index. The Generator component uses an LLM such as GPT, LLaMA, or Mistral-7B.

5. Inference frameworks and deployment

PyTorch is a Framework used for model training and experimentation. TensorFlow is a Framework used for machine learning. ONNX Runtime is a Framework used for model inference. vLLM is a Framework used for high-throughput LLM serving.

vLLM serves LLaMA, Mistral-7B, Mixtral-8x7B, and Qwen. ONNX Runtime runs optimized neural networks. Hugging Face Transformers is a Framework used to load BERT, GPT, T5, BART, and many other models.

DeepSpeed is a Framework used for distributed training. DeepSpeed uses ZeRO optimization. ZeRO is a Method used to reduce memory consumption during large model training.

6. Explicit relation sentences for testing

Expected semantic relations:

BERT USES Masked Language Modeling.

BERT USED_FOR Named Entity Recognition.

BERT USED_FOR Text Classification.

GPT USED_FOR Text Generation.

GPT USED_FOR Summarization.

LLaMA USES Rotary Positional Embeddings.

LLaMA USES SwiGLU.

Mixtral-8x7B USES Mixture of Experts.

CLIP USES Contrastive Learning.

CLIP RELATED_TO Joint Embedding Space.

RAG USES Vector Index.

RAG USED_FOR Question Answering.

Milvus USED_FOR Vector Search.

Qdrant USED_FOR Vector Search.

Weaviate USED_FOR Vector Search.

vLLM SERVES LLaMA.

vLLM SERVES Mistral-7B.

DeepSpeed USES ZeRO.

LoRA USED_FOR Fine-tuning.

QLoRA USES Quantization.

7. Short control paragraph

If the pipeline works correctly, the final graph should contain clean nodes such as BERT, GPT, LLaMA, Mistral-7B, Mixtral-8x7B, CLIP, RAG, LoRA, QLoRA, Milvus, Qdrant, Weaviate, LlamaIndex, LangChain, PyTorch, vLLM, Vector Index, Embedding, Self-Attention, and Masked Language Modeling.

The graph should not create nodes from ordinary words such as document, paragraph, example, section, source, component, or pipeline unless these words are part of a meaningful technical term.